

AGENTS.md — Spot-Welder multi-firmware workspace

Guidance for AI coding agents working in this repo. Read this before editing.

Repository map

This is a **mono-repo** containing multiple independent firmware/software projects that communicate over a shared serial/TCP telemetry protocol.

Path	Target	Toolchain	Role
ESP32P4/	ESP32-P4 (CrowPanel Advance 5")	ESP-IDF (CMake)	UI + WiFi bridge + STM32/OTA flasher. Talks to STM32 over UART3-IN @ 576000. Enriches STATUS, re-lays WAVEFORM_* raw.
STM32G474CE/	STM32G474CE	PlatformIO (env:g474ceu6_stlink)	Real-time weld controller: FET drive, ADC/shunt, INA226, charger, joule integrator. Sole producer of STATUS, WELD_DONE, WAVEFORM_* packets.
ESP32_8048S043C/	ESP32-8048S043C (old board)	PlatformIO / Arduino	Legacy display firmware. Produces DISPLAY and CELLS packets. Keep buildable; not the primary target.
Spotwelder Full/	(build artifacts only)	(no source)	Not an active firmware target. Contains only .pio/ , build.log , sdkconfig.defaults .
(separate repo) Spot-Welder-Server/	Python 3 / Flask	pip	Dashboard + TCP client to the P4 bridge (:8888). Re-parses telemetry.

The Flask server lives in a **separate repository** (Spot-Welder-Server). For any change that touches the wire protocol, that repo **MUST** be edited in the same coordinated change.



Source of truth: the telemetry protocol

There is currently **no shared protocol header**. The STATUS / WAVEFORM_DATA packet is:

- **Emitted** in STM32G474CE/src/main.c (search "STATUS,") as an ASCII key=value CSV line.
- **Parsed** in ESP32P4/main/wifi_bridge.cpp and welder_main.cpp (sscanf / strtok).
- **Parsed again** in Spot-Welder-Server/app.py .

See `PROTOCOL.md` at the repo root for a descriptive reference of observed packet formats.

Rule: any change to a field name, order, type, or units is a breaking change across all three.

When editing the protocol, update ALL of: the STM32 emitter, both P4 parsers, and the Flask parser — and keep this list in sync. Prefer additive changes (append new fields) over reordering.

Build & validation commands

Always build the project(s) you touched before claiming a change is done.

ESP32-P4 (ESP-IDF)

```
cd ESP32P4
idf.py set-target esp32p4      # first time only
idf.py build                  # compile – MUST pass with no errors
idf.py -p <PORT> flash monitor # flash + serial (hardware only)
```

- Output binary: `build/esp32_firmware.bin` (project name is `esp32_firmware`).
- Do NOT flash `hello_world.bin` — that name is stale/removed.

STM32G474CE (PlatformIO)

```
cd STM32G474CE
pio run                      # compile env:g474ceu6_stlink – MUST pass
pio run -t upload            # flash via ST-Link (hardware only)
pio device monitor -b 576000 # serial monitor at app baud
```

Legacy ESP32-8048S043C (PlatformIO)

```
cd ESP32_8048S043C
pio run                      # keep it compiling; secondary priority
```

Note: `Spotwelder Full/` contains only build artifacts and is not a build target.

Flask server (separate repo)

```
cd Spot-Welder-Server
pip install -r requirements.txt
python app.py
# serves dashboard on :8080, TCP bridge client to P4 :8888
```

Hardware / safety constraints the agent MUST respect

- **UART link is 576000 8N1** through BSS138 level shifters on the UART3-IN header (GPIO 27/28 on P4, PA9/PA10 on STM32). Do NOT raise the baud rate — higher rates corrupt through the RC-limited shifters. Bootloader mode uses 115200 8E1.
- **The charge MOSFET (`CHARGER_EN`) is owned by the STM32** and MUST be OFF during a weld pulse — the joule integrator assumes discharge-only current. Never move this control to the P4.

- **Reserved:** do not bind host ports 1000/2200 in any tooling.

Conventions

- C/C++: match existing style in each project (no reformatting unrelated code).
- Guard all float telemetry math with `isfinite() + a < 0` floor, as the STM32 code already does.
- Never commit, push, open a PR, merge, or auto-merge unless I explicitly request it. When changes are complete, provide a per-repository diff summary.
- After any protocol edit, state explicitly which of the 3 parsers you updated.